

# Nota 04 – Ejemplos con TikZ

Graficas, vectores y figuras geometricas

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## Índice

## 1. Que es TikZ

TikZ es un paquete de LaTeX que permite crear graficas, diagramas y figuras geometricas directamente en el documento.

### Tip

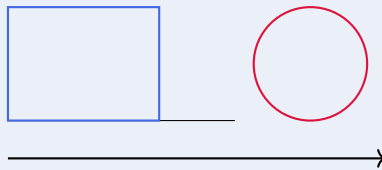
Paquetes necesarios en el preambulo:

```
\usepackage{tikz}
\usepackage{pgfplots}
\pgfplotsset{compat=1.18}
```

## 2. Figuras geometricas basicas

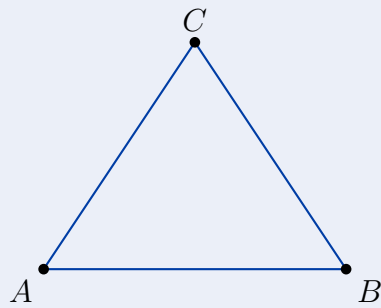
### Lineas-rectangulos-circulos

```
\begin{tikzpicture}
  \draw (0,0) -- (3,0);
  \draw[blue, thick] (0,0) rectangle (2,1.5);
  \draw[red, thick] (4,0.75) circle (0.75);
  \draw[->, thick] (0,-0.5) -- (5,-0.5);
\end{tikzpicture}
```



## Triangulo con etiquetas

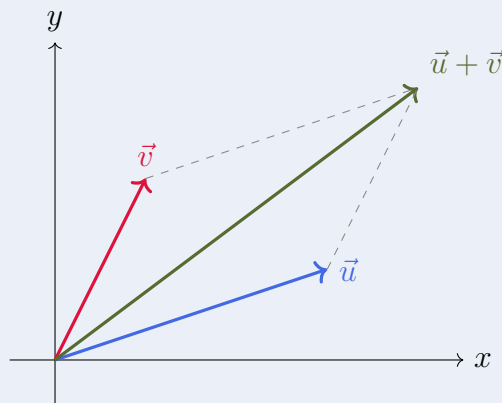
```
\begin{tikzpicture}
  \coordinate (A) at (0,0);
  \coordinate (B) at (4,0);
  \coordinate (C) at (2,3);
  \draw[thick] (A) -- (B) -- (C) -- cycle;
  \node[below left] at (A) {$A$};
  \node[below right] at (B) {$B$};
  \node[above] at (C) {$C$};
  \fill (A) circle (2pt);
  \fill (B) circle (2pt);
  \fill (C) circle (2pt);
\end{tikzpicture}
```



### 3. Vectores

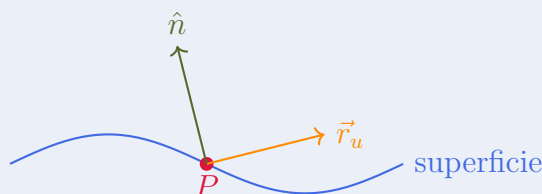
#### Suma de vectores

```
\begin{tikzpicture}[scale=1.2]
  \draw[->] (-0.5,0) -- (4.5,0) node[right] {$x$};
  \draw[->] (0,-0.5) -- (0,3.5) node[above] {$y$};
  \draw[->, very thick, miAzul]
    (0,0) -- (3,1) node[right] {$\vec{u}$};
  \draw[->, very thick, miRojo]
    (0,0) -- (1,2) node[above] {$\vec{v}$};
  \draw[->, very thick, miVerde]
    (0,0) -- (4,3) node[above right] {$\vec{u}+\vec{v}$};
  \draw[miGris, dashed] (3,1) -- (4,3);
  \draw[miGris, dashed] (1,2) -- (4,3);
\end{tikzpicture}
```



#### Vector normal a una superficie

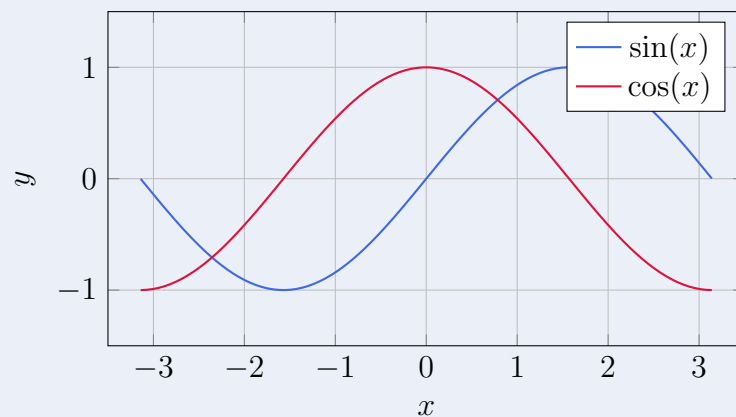
```
\begin{tikzpicture}[scale=1.3]
  \draw[thick, miAzul, domain=0:4, samples=50]
    plot (\x, {0.3*sin(180*\x/2)}) node[right] {superficie};
  \coordinate (P) at (2, 0);
  \fill[miRojo] (P) circle (2pt) node[below] {$P$};
  \draw[->, thick, miNaranja]
    (P) -- ++(1.2, 0.3) node[right] {$\vec{r}_u$};
  \draw[->, thick, miVerde]
    (P) -- ++(-0.3, 1.2) node[above] {$\hat{n}$};
\end{tikzpicture}
```



## 4. Graficas de funciones con pgfplots

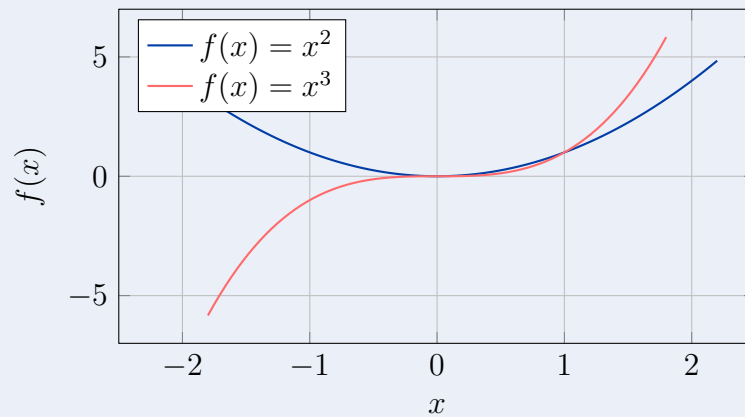
### Seno y coseno

```
\begin{tikzpicture}
\begin{axis}[
width=10cm, height=6cm,
xlabel={x}, ylabel={y},
xmin=-3.5, xmax=3.5,
ymin=-1.5, ymax=1.5,
grid=both, legend pos=north east,
]
\addplot[miAzul, thick, domain=-3.14:3.14, samples=100]
{sin(deg(x))};
\addlegendentry{$\sin(x)$}
\addplot[miRojo, thick, domain=-3.14:3.14, samples=100]
{cos(deg(x))};
\addlegendentry{$\cos(x)$}
\end{axis}
\end{tikzpicture}
```



## Parabola y cubica

```
\begin{tikzpicture}
\begin{axis}[
width=10cm, height=6cm,
xlabel={x}, ylabel={f(x)},
xmin=-2.5, xmax=2.5,
grid=both, legend pos=north west,
]
\addplot[azulIPn, thick, domain=-2.2:2.2, samples=80]{x^2};
\addlegendentry{$f(x)=x^2$}
\addplot[rojoSuper, thick, domain=-1.8:1.8, samples=80]{x^3};
\addlegendentry{$f(x)=x^3$}
\end{axis}
\end{tikzpicture}
```



## 5. Comandos TikZ mas usados

| Comando                         | Para que sirve           |
|---------------------------------|--------------------------|
| <code>\draw</code>              | dibujar lineas y figuras |
| <code>\fill</code>              | rellenar con color       |
| <code>\node</code>              | agregar texto            |
| <code>\coordinate</code>        | definir punto con nombre |
| <code>--</code>                 | linea recta              |
| <code>cycle</code>              | cerrar figura            |
| <code>-&gt;</code>              | flecha al final          |
| <code>domain=a:b</code>         | rango de funcion         |
| <code>samples=n</code>          | puntos a calcular        |
| <code>plot (\x, {f(\x)})</code> | graficar funcion         |